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Section 4-5:15pm M/W
Written Analysis

Question 1)

Formula - $h(n)$ = number of positions where the current state differs from goal

For the steps cost, my heuristic is the distance between my current and goal word, which means in the number of places the two words differ. **Unavoidable cost:** As each action can only change one character, every mismatched character must be changed at least once before getting to the goal. Therefore, if there are n mismatches, then there is at least n steps would be required. With this heuristic returning the absolute minimum amount of changes, it cannot overestimate the actual remaining steps and is admissible.

Formula - $h(n)$ - sum of the scrabble values of the goal letters at every position where the current state differs from goal

For the scrabble cost, my heuristic sums the scrabble letters of the goal word in each letter which differs from the original word. **Unavoidable cost:** Every mismatched position has to be changed, so that the correct goal letter is put into the right place/position. Since every goal letter that has to be inserted has the associated scrabble cost, the sum of those goal letters costs is the minimum unavoidable amount. If there are intermediate moves between the start and goal state, that would only increase the true remaining cost.

Therefore, this heuristic does not overestimate and is admissible as it only has the minimum required cost calculated for it.

Formula - $h(n)$ = number of positions where the current state differs from goal

For the frequency cost, my heuristic also does the same counting the number of mismatched positions (letters) between the original word and the goal word. **Unavoidable cost:** So for every mismatch, we know the cost will be at least 1 as we know the formula is $1 + \text{rarity}(\text{word})$, and the rarity value is nonnegative. Therefore if there are n mismatches, there will be a true cost of at least n .

The heuristic returns n , so it therefore cannot overestimate and is admissible.

Question 2)

Word pair	Cost measure	UCS cost	UCS nodes expanded	A* cost	A* nodes expanded	Reduction %
WHY → ASK	Steps	5	773	5	16	97.93
WHY → ASK	Scrabble	9	765	9	13	98.3

WHY → ASK	Frequency	38.85540652	808	38.85540652	755	6.56
BEAR → DUCK	Steps	4	691	4	6	99.13
BEAR → DUCK	Scrabble	11	2981	11	6	99.80
BEAR → DUCK	Frequency	38.2402284	1084	38.2402284	720	33.58

- Yes, both A* and UCS always returned the same costs. For WHY → ASK, the costs were as follows: Steps:5, Scrabble:9, Frequency: 38.85540652. For BEAR → DUCK the costs were as follows: Steps:4, Scrabble:11, Frequency: 38.2402284. Both can be seen in the above chart
- The largest reduction for A* for the WHY → ASK, was scrabble at 98.3%, and for BEAR → DUCK it was also scrabble at 99.80%
- For both word pairs (WHY → ASK and BEAR → DUCK), the least informative heuristic was the frequency heuristic. Its reductions were only 6.56% and 33.58%, which were much smaller reductions in comparison to step and scrabble.
- The steps heuristic is informative as each mismatched letter must be replaced at least once, while the scrabble is more specific as it adds the scrabble values of the goal letters which still need to be inserted. The frequency heuristic only counts the mismatched letters and does not consider the rarity of the word that may need to be entered. As frequency costs can vary depending on the word's rarity, it gives A* less information about the true remaining cost, which made it expand a lot more nodes for the frequency cases.

3)

For **steps**, the answer is yes, as every mismatched letter adds one to the cost, so the cost would be the same in either direction.

For **scrabble**, the answer is no as each letter that has a new letter that needs to take its place. Each letter has its own associated scrabble value cost and different letters have different costs, so going from word 1 to word 2 would have different cost than going from word 2 to word 1, as different goal letters are needed with different costs.

An example would be dog → cog, as cost(c) = 3, while for cog → dog, cost(d)=2, showing that reversibility and the symmetry of cost would not be the same.

For **frequency**, the answer is no as each word has its own rarity. Reversing enters a different word, and therefore the cost would change based on its rarity. In the case of cog → dog, it would be 1 + rarity(dog) which is 1 + 5.819885 = 6.819885 and for dog → cog, it would be 1 +

rarity(cog) which is $1 + 11.393166 = 12.393166$. This shows that an action can be reversible even when its cost is not symmetric.

4) Prompt:

Model: Chat-GPT 5.6 sol

In DOGCAT, the initial and goal states are lowercase words of the same length—three or four letters—and every legal state must appear in the supplied dictionary. A legal action changes exactly one letter. Under the frequency objective, entering word w costs $1 + \text{rarity}(w)$, where the nonnegative rarity value is stored in the dictionary.

The baseline heuristic $h_0(n)$ is the number of positions in which state n differs from the goal. Propose a rarity-aware heuristic intended to be more informative than h_0 while remaining admissible. Define it precisely and justify your proposal.

By “more informative,” we mean that the proposed heuristic should be at least as large as h_0 for every state and strictly larger for at least one state, without overestimating the true optimal remaining cost.

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Proposed Formula:

$$h(n) = \begin{cases} 0, & n = \text{goal} \\ d + \sum_{k=0}^{d-1} r_k, & \text{otherwise.} \end{cases}$$

3) I believe the proposed heuristic is admissible. As a legal action only changes one letter, so the number of mismatched positions between the current word and the goal can only change by at most once per move. Therefore, if the current state is d mismatches away, every solution will pass by the $d-1$, $d-2$, ..., mismatch levels with a cost of at least 1, which gives the base cost of d . When a path enters distance level k , it would still have to enter some dictionary word at that level, and the rarity must be at least r_k as r_k is the minimum rarity of any word at that level. For example if it is $d-2$ (2 mismatches away), it will look at all the legal dictionary words that are 2 mismatches from the goal word and add the smallest rarity to the amount. Therefore, the heuristic is adding the cost that every solution must pay, so it cannot overestimate the true cost.

4) Not applicable

5) I tested the AI proposed rarity aware heuristic against the baseline frequency heuristic on the two required word pair problems: WHY $\rightarrow$ ASK and BEAR $\rightarrow$ DUCK. For WHY $\rightarrow$ ASK, the

baseline heuristic returned the optimal cost 38.85540652 and expanded 755 nodes, while the rarity aware heuristic returned the same optimal cost but only expanded 294 nodes.

For BEAR → DUCK, the baseline heuristic returned the optimal cost 38.2402284 and expanded 720 nodes, while the rarity aware heuristic returned the same optimal cost but expanded only 75 nodes. This is a reduction of 61.06% for WHY → ASK and 89.58% for BEAR → DUCK. Since both heuristics returned the same optimal cost but the rarity aware heuristic expanded fewer nodes, the evidence supports using the AI proposed heuristic over the baseline one.